ADITYA KUMAR SINGH

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EDUCATION

Madhav Institute of Technology & Science — Gwalior, India
B. Tech in Artificial Intelligence and Machine Learning — *Aug. 2023 – Present*

EXPERIENCE

AI Development Intern | Zipbolt Innovations — *Sep. 2025 – Jan. 2026*

- Worked on **ZipCRM**, an internal CRM platform built using the **PERN stack (PostgreSQL, Express, React, Node.js)**, contributing to backend APIs, database design, and workflow automation.
- Built **AI orchestration pipelines** for CRM automation, including data ingestion, processing, and task execution, improving operational efficiency and reducing manual intervention.
- Developed a **full-stack EV rental management system**, handling backend services, business logic, and database schema while integrating AI-powered decision workflows.
- Implemented **event-driven backend and AI orchestration using Inngest**, enabling scalable background jobs, async task execution, and intelligent automation inside the CRM.
- Collaborated closely with product and engineering teams to translate business requirements into scalable backend systems and production-ready AI solutions.

Content Creator | Connect With Aadi — *Jan. 2025 – Present*

- Built a personal brand focused on AI, technology, and entrepreneurship, creating engaging content on Instagram and YouTube.
- Produced AI-driven educational reels and tutorials, simplifying complex concepts for a wider audience.
- Implemented growth strategies (HBC formula, data-driven content hooks), resulting in [X% increase in followers/views].
- Collaborated with peers on real-world problems, turning insights into practical, relatable content for the community.
- Strengthened skills in AI research, digital marketing, storytelling, and audience engagement.

PROJECTS

Legal RAG Assistant

- **Tech Stack:** Python, LangChain, FAISS, OpenAI Embeddings, PyMuPDF, Streamlit, AutoGen
- Built an end-to-end Retrieval-Augmented Generation (RAG) system for legal document analysis and question answering
- Implemented document ingestion and parsing for PDFs using PyMuPDF, with optimized text chunking strategies
- Developed semantic search using FAISS vector database for efficient retrieval of relevant document context
- Integrated LLMs to generate context-aware and accurate responses based on retrieved information
- Designed an interactive UI using Streamlit for real-time document querying
- Explored multi-agent workflows (AutoGen) to enhance reasoning and response generation

Decoder-Only Transformer (GPT-style LLM) from Scratch

- **Tech Stack:** Python, PyTorch, NumPy, Transformers (for Tokenizers), Jupyter Notebook
- Implemented the core architecture of a transformer model from the ground up, including multi-head self-attention, positional encoding, and layer normalization.
- Developed a complete training pipeline for text generation, handling data pre-processing, BPE tokenization, model training loop, and inference.
- Analyzed model performance by tracking training/validation loss curves and evaluating the quality of generated text to understand training dynamics.
- Meticulously documented the entire codebase and architectural decisions in a detailed GitHub repository to ensure clarity and reproducibility.

Technical Skills

Programming: Python, Java, C, C++, JavaScript, SQL, HTML/CSS, R

AI/ML & Data Science: TensorFlow, PyTorch, scikit-learn, NumPy, Pandas, Matplotlib, NLP, Deep Learning

GenAI: LangChain, crewAI, Autogen, Ollama, n8n, crewAI

Research & Analytics: Market Research, Data Analysis (Excel, Google Sheets, PowerBI), Statistics & Visualization (regression, charts/graphs)

Certifications

- Complete Data Science,Machine Learning,DL,NLP Bootcamp
- AI in Social Media SIP, MITS, Gwalior -Dec 2024